Hackthon YRSB Analysis

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Background

This analysis investigates the association between **sexual identity** and **suicidal ideation (SI)** among U.S. high school students using the **2023 Youth Risk Behavior Survey (YRBS)**.

Research Question: > Is sexual identity associated with suicidal ideation? Does race/ethnicity confound or modify this association?

Variables:

Role	Variable	Type
Exposure	Sexual Identity	Binary (Heterosexual vs Non-Het/Questioning)
Outcome	Suicidal Ideation	Binary (Yes/No)
Confounder/EMM	Race/Ethnicity	Categorical (8 groups)
Covariates	Age, Sex, Low Grades, Bullying	Binary/Categorical

```r
library(readr)
library(dplyr)
library(ggplot2)
library(tableone)
library(table1)
```

```
## [1] "/Users/tejsharm/Downloads"
```

```r
data <- readRDS("/Users/tejsharm/Downloads/yrbs2023_final.rds")
attach(data)

cat("Dataset dimension:", nrow(data),"rows x", ncol(data), "columns\n")
```

```
## Dataset dimension: 20103 rows x 29 columns
```

```r
# Explore Key Variables (Before Cleaning)
# Look at raw distribution of key variables before any recoding
cat("=== Sexual Identity (raw) ===\n")
```

```
## === Sexual Identity (raw) ===
```

```r
table(data$sexual_identity, useNA = "always")
```

```
##
##     1     2     3     4     5     6  <NA>
## 13289   683  2053   760   850   465  2003
```

```r
cat("\n=== Suicidal Ideation (raw) ===\n")
```

```
##
## === Suicidal Ideation (raw) ===
```

```r
table(data$suicideideation_binary, useNA = "always")
```

```
##
##     0     1  <NA>
## 15453  4214   436
```

```r
cat("\n=== Age (raw) ===\n")
```

```
##
## === Age (raw) ===
```

```r
table(data$age, useNA = "always")
```

```
##
##    1    2    3    4    5    6    7 <NA>
##   44   33 2569 5526 5208 4458 2167   98
```

```r
cat("\n=== Race/Ethnicity (raw) ===\n")
```

```
##
## === Race/Ethnicity (raw) ===
```

```r
table(data$raceeth, useNA = "always")
```

```
##
##    1    2    3    4    5    6    7    8 <NA>
## 1334  995 1791  105 9700 1208 2786 1814  370
```

```r
table1 <- table(sexual_identity, suicideideation_binary)

pct_table1 <- round(prop.table(table1, 2) *100, 2)

pct_table1
```

```
##                 suicideideation_binary
## sexual_identity     0     1
##               1 79.85 50.20
##               2  2.97  6.48
##               3  7.72 24.62
##               4  2.90  8.91
##               5  3.80  7.97
##               6  2.75  1.82
```

```r
table(raceeth, suicideideation_binary)
```

```
##        suicideideation_binary
## raceeth    0    1
##       1 1002  307
##       2  822  154
##       3 1414  339
##       4   84   20
##       5 7475 2072
##       6 1010  167
##       7 2126  624
##       8 1304  471
```

```r
table(age, suicideideation_binary)
```

```
##    suicideideation_binary
## age    0    1
##   1   16   24
##   2   27    4
##   3 1983  524
##   4 4278 1125
##   5 3962 1141
##   6 3473  907
##   7 1638  476
```

```r
# Data Cleaning
table1_clean <- data %>%
  dplyr::select(sexual_identity, suicideideation_binary,
                raceeth, age, bulliedschool_binary,
                sex, lowgrades_binary) %>%

  filter(age %in% c("3","4","5","6","7")) %>%

  mutate(
    sexual_identity_cat = case_when(
      sexual_identity == "1"                  ~ 0,
      sexual_identity %in% c("2","3","4","5") ~ 1,
      TRUE                                    ~ NA_real_),

    SI = case_when(
      suicideideation_binary == "1" ~ 1,
      suicideideation_binary == "0" ~ 0,
      TRUE                          ~ NA_real_),
```

```r
    sex_clean = case_when(
      sex == "1" ~ 1,
      sex == "0" ~ 0,
      TRUE        ~ NA_real_),

    lowgrades_clean = case_when(
      lowgrades_binary == "1" ~ 1,
      lowgrades_binary == "0" ~ 0,
      TRUE                    ~ NA_real_),

    bullied_clean = case_when(
      bulliedschool_binary == "1" ~ 1,
      bulliedschool_binary == "0" ~ 0,
      TRUE                        ~ NA_real_)
  )


cat("Columns in clean data:\n")
```

```
## Columns in clean data:
```

```r
names(table1_clean)
```

```
##  [1] "sexual_identity"        "suicideideation_binary" "raceeth"
##  [4] "age"                    "bulliedschool_binary"   "sex"
##  [7] "lowgrades_binary"       "sexual_identity_cat"    "SI"
## [10] "sex_clean"              "lowgrades_clean"        "bullied_clean"
```

Descriptive Statistics (Table 1)

Table 1 describes the characteristics of the study population, both overall and stratified by sexual identity (the exposure). This helps us see whether the two groups (heterosexual vs non-heterosexual) differ on key covariates — which tells us which variables we may need to control for.

```r
# Remove NAs in exposure variable for stratified table only
# (We only need to know which group each student is in for the stratified table)
yrbs_mod <- table1_clean %>% filter(!is.na(sexual_identity_cat))

# Overall Table 1 - describes the full sample
table1(~ factor(SI) + factor(sex_clean) + factor(age) +
         factor(raceeth) + factor(lowgrades_clean) +
         factor(bullied_clean),
       data = table1_clean)
```

	Overall
	(N=19928)

	Overall
factor(SI)	
0	15334 (76.9%)
1	4173 (20.9%)
Missing	421 (2.1%)
factor(sex_clean)	
0	9973 (50.0%)
1	9816 (49.3%)
Missing	139 (0.7%)
factor(age)	
3	2569 (12.9%)
4	5526 (27.7%)
5	5208 (26.1%)
6	4458 (22.4%)
7	2167 (10.9%)
factor(raceeth)	
1	1327 (6.7%)
2	984 (4.9%)
3	1782 (8.9%)
4	101 (0.5%)
5	9632 (48.3%)
6	1200 (6.0%)
7	2752 (13.8%)
8	1800 (9.0%)
Missing	350 (1.8%)
factor(lowgrades_clean)	
0	11587 (58.1%)
1	4817 (24.2%)
Missing	3524 (17.7%)
factor(bullied_clean)	
0	15742 (79.0%)
1	4001 (20.1%)
Missing	185 (0.9%)

```r
# Stratified Table 1 - compares the two exposure groups side by side
# This answers: "Are heterosexual and non-het students different from each other?"
table1(~ factor(SI) + factor(sex_clean) + factor(age) +
         factor(raceeth) + factor(lowgrades_clean) +
         factor(bullied_clean) | sexual_identity_cat,
       data = yrbs_mod)
```

	0	1	Overall
	(N=13196)	(N=4309)	(N=17505)
factor(SI)			
0	11214 (85.0%)	2438 (56.6%)	13652 (78.0%)
1	1865 (14.1%)	1778 (41.3%)	3643 (20.8%)
Missing	117 (0.9%)	93 (2.2%)	210 (1.2%)
factor(sex_clean)			
0	7652 (58.0%)	1066 (24.7%)	8718 (49.8%)
1	5519 (41.8%)	3170 (73.6%)	8689 (49.6%)
Missing	25 (0.2%)	73 (1.7%)	98 (0.6%)

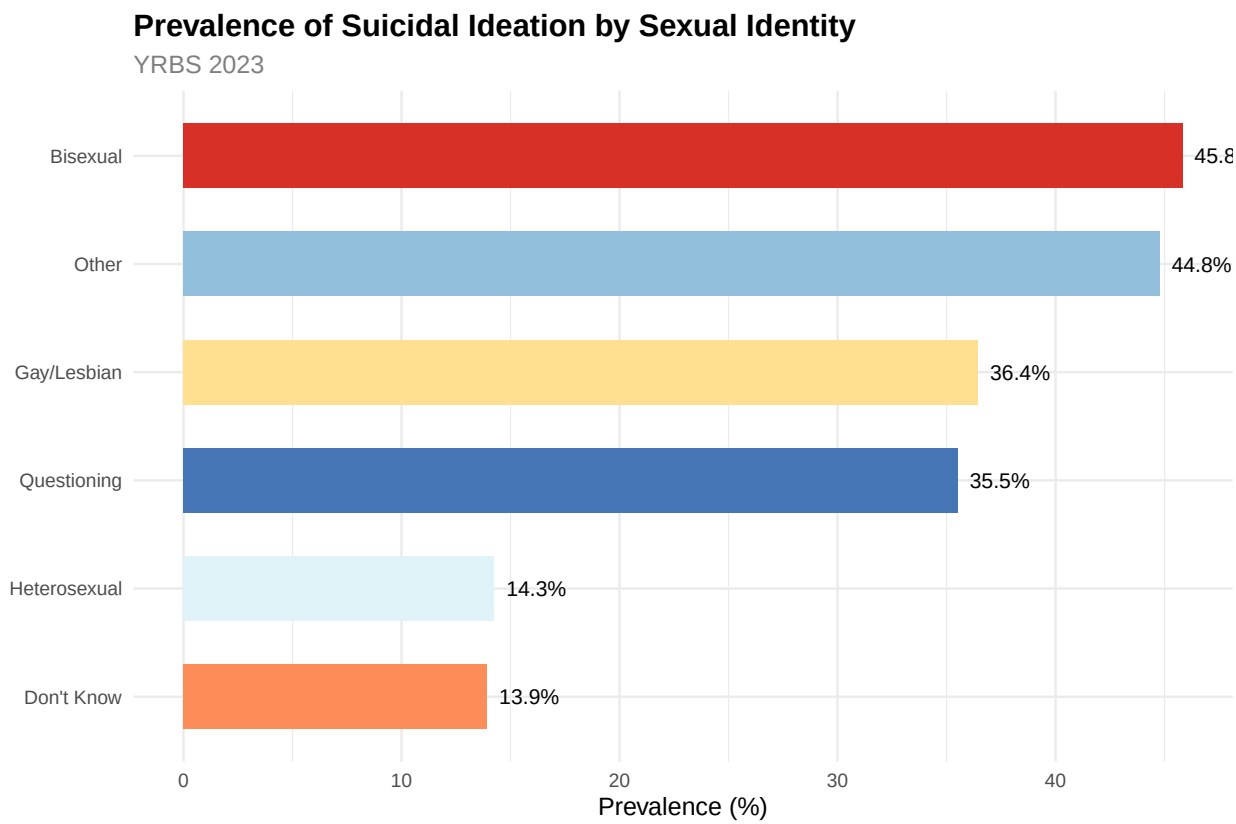
	0	1	Overall
factor(age)			
3	1687 (12.8%)	552 (12.8%)	2239 (12.8%)
4	3691 (28.0%)	1165 (27.0%)	4856 (27.7%)
5	3430 (26.0%)	1159 (26.9%)	4589 (26.2%)
6	2964 (22.5%)	987 (22.9%)	3951 (22.6%)
7	1424 (10.8%)	446 (10.4%)	1870 (10.7%)
factor(raceeth)			
1	734 (5.6%)	221 (5.1%)	955 (5.5%)
2	679 (5.1%)	202 (4.7%)	881 (5.0%)
3	1205 (9.1%)	432 (10.0%)	1637 (9.4%)
4	69 (0.5%)	13 (0.3%)	82 (0.5%)
5	6585 (49.9%)	2081 (48.3%)	8666 (49.5%)
6	871 (6.6%)	206 (4.8%)	1077 (6.2%)
7	1814 (13.7%)	641 (14.9%)	2455 (14.0%)
8	1108 (8.4%)	447 (10.4%)	1555 (8.9%)
Missing	131 (1.0%)	66 (1.5%)	197 (1.1%)
factor(lowgrades_clean)			
0	8020 (60.8%)	2357 (54.7%)	10377 (59.3%)
1	3091 (23.4%)	1224 (28.4%)	4315 (24.7%)
Missing	2085 (15.8%)	728 (16.9%)	2813 (16.1%)
factor(bullied_clean)			
0	10942 (82.9%)	2987 (69.3%)	13929 (79.6%)
1	2194 (16.6%)	1292 (30.0%)	3486 (19.9%)
Missing	60 (0.5%)	30 (0.7%)	90 (0.5%)

SI Prevalence by Sexual Identity Group (All 6 Categories)

Before collapsing to binary, it is useful to visualize SI prevalence across ALL sexual identity categories to understand the full picture.

```
# Calculate prevalence
# Easy to understand, and good for presentation
prev_data <- table1_clean %>%
  filter(!is.na(sexual_identity), !is.na(SI)) %>%
  mutate(si_label = recode(as.character(sexual_identity),
    "1" = "Heterosexual",
    "2" = "Gay/Lesbian",
    "3" = "Bisexual",
    "4" = "Other",
    "5" = "Questioning",
    "6" = "Don't Know")) %>%
  group_by(si_label) %>%
  summarise(prevalence = mean(SI, na.rm = TRUE) * 100,
            n = n()
            )

ggplot(prev_data, aes(x = reorder(si_label, prevalence),
                      y = prevalence,
```

```r
                      fill = si_label)) +
  geom_bar(stat = "identity", width = 0.6) +
  geom_text(aes(label = paste0(round(prevalence, 1), "%")),
            hjust = -0.2, size = 4) +
  coord_flip() +
  scale_fill_brewer(palette = "RdYlBu") +
  labs(title    = "Prevalence of Suicidal Ideation by Sexual Identity",
       subtitle = "YRBS 2023",
       x        = NULL,
       y        = "Prevalence (%)") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none",
        plot.title      = element_text(face = "bold"),
    plot.subtitle   = element_text(color = "grey50"),
    plot.caption    = element_text(color = "grey50"))
```



Association Testing

Chi-Square Test

The chi-square test asks: **"Are sexual identity and suicidal ideation statistically associated?"** It tests whether the distribution of SI differs between heterosexual and non-heterosexual students.

```r
# First make a cross table with readable label
cross_tab <- table(table1_clean$sexual_identity_cat,
                   table1_clean$SI)

# Add labels so it's readable
dimnames(cross_tab) <- list(
  "Sexual Identity"   = c("Heterosexual", "Non-Het/Questioning"),
  "Suicidal Ideation" = c("No", "Yes")
)

# See the raw counts
print(cross_tab)
```

```
##                      Suicidal Ideation
## Sexual Identity          No   Yes
##   Heterosexual        11214  1865
##   Non-Het/Questioning  2438  1778
```

```r
prop.table(cross_tab, margin = 1) * 100
```

```
##                      Suicidal Ideation
## Sexual Identity             No      Yes
##   Heterosexual        85.74050 14.25950
##   Non-Het/Questioning 57.82732 42.17268
```

```r
chisq.test(cross_tab, correct = FALSE)
```

```
##
##  Pearson's Chi-squared test
##
## data:  cross_tab
## X-squared = 1494, df = 1, p-value < 2.2e-16
```

```r
# p < 0.05 = statistically significant association
```

Crude (Unadjusted) Odds Ratio

The crude OR tells us the **raw strength of association** between sexual identity and SI, before accounting for any other variables. It answers: "How much higher are the odds of SI in non-het students compared to heterosexual students?"

```r
model_crude <- glm(SI ~ sexual_identity_cat,
                   data   = table1_clean,
                   family = binomial(link = "logit"))

summary(model_crude)
```

```
##
## Call:
## glm(formula = SI ~ sexual_identity_cat, family = binomial(link = "logit"),
##     data = table1_clean)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -1.79390    0.02501  -71.73   <2e-16 ***
## sexual_identity_cat  1.47821    0.03997   36.98   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 17807  on 17294  degrees of freedom
## Residual deviance: 16456  on 17293  degrees of freedom
##   (2633 observations deleted due to missingness)
## AIC: 16460
##
## Number of Fisher Scoring iterations: 4
```

```r
cat("\n=== Crude OR with 95% CI ===\n")
```

```
##
## === Crude OR with 95% CI ===
```

```r
round(exp(cbind(OR = coef(model_crude),
                confint.default(model_crude))), 2)
```

```
##                       OR 2.5 % 97.5 %
## (Intercept)         0.17  0.16   0.17
## sexual_identity_cat 4.39  4.05   4.74
```

Adjusted Logistic Regression

Why adjust? The crude OR may be distorted by other variables (confounders). For example, non-het students may be more likely to get bullied, and bullying independently increases SI risk. By adjusting, we isolate the **true** effect of sexual identity.

Why `relevel()` for raceeth and age only? These are the only variables with 3+ categories. Binary variables (sex, grades, bullying) automatically use 0 as the reference — no manual specification needed.

Adjustd OR + 95% CI

This code converts the "raw" math of the model into something human can easily understand: Odds Rations.

If you run logistic regression (glm), the model output is in Log-Odds.

Since humans don't thinkin in Log-Odds, we use exp() to convert them into odd ratio (OR), which is easier to interpret.

```r
model_adjusted <- glm(SI ~ sexual_identity_cat +
                          relevel(factor(raceeth), ref = "5") +
                          relevel(factor(age),     ref = "5") +
                          sex_clean +
                          lowgrades_clean +
                          bullied_clean,
                      data   = table1_clean,
                      family = binomial(link = "logit"))

summary(model_adjusted)
```

```
##
## Call:
## glm(formula = SI ~ sexual_identity_cat + relevel(factor(raceeth),
##     ref = "5") + relevel(factor(age), ref = "5") + sex_clean +
##     lowgrades_clean + bullied_clean, family = binomial(link = "logit"),
##     data = table1_clean)
##
## Coefficients:
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                         -2.364241   0.061256 -38.596  < 2e-16 ***
## sexual_identity_cat                  1.210272   0.047683  25.381  < 2e-16 ***
## relevel(factor(raceeth), ref = "5")1  0.065348   0.096604   0.676 0.498751
## relevel(factor(raceeth), ref = "5")2 -0.241758   0.115660  -2.090 0.036596 *
## relevel(factor(raceeth), ref = "5")3 -0.110810   0.083108  -1.333 0.182424
## relevel(factor(raceeth), ref = "5")4 -0.037586   0.319176  -0.118 0.906258
## relevel(factor(raceeth), ref = "5")6 -0.409971   0.105748  -3.877 0.000106 ***
## relevel(factor(raceeth), ref = "5")7  0.008741   0.065403   0.134 0.893685
## relevel(factor(raceeth), ref = "5")8  0.276556   0.074943   3.690 0.000224 ***
## relevel(factor(age), ref = "5")3     -0.197382   0.078946  -2.500 0.012411 *
## relevel(factor(age), ref = "5")4     -0.160247   0.061238  -2.617 0.008876 **
## relevel(factor(age), ref = "5")6     -0.036663   0.063218  -0.580 0.561946
## relevel(factor(age), ref = "5")7      0.138478   0.076818   1.803 0.071438 .
## sex_clean                             0.475323   0.047442  10.019  < 2e-16 ***
## lowgrades_clean                       0.557477   0.048157  11.576  < 2e-16 ***
## bullied_clean                         1.135985   0.049207  23.086  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 14722  on 14269  degrees of freedom
## Residual deviance: 12775  on 14254  degrees of freedom
##   (5658 observations deleted due to missingness)
## AIC: 12807
##
## Number of Fisher Scoring iterations: 4
```

```r
cat("\n--Adjusted OR with 95% CI --\n")
```

```
##
## --Adjusted OR with 95% CI --
```

```r
round(exp(cbind(OR = coef(model_adjusted),
                confint.default(model_adjusted))), 2)
```

```
##                                        OR 2.5 % 97.5 %
## (Intercept)                          0.09  0.08   0.11
## sexual_identity_cat                  3.35  3.06   3.68
## relevel(factor(raceeth), ref = "5")1 1.07  0.88   1.29
## relevel(factor(raceeth), ref = "5")2 0.79  0.63   0.99
## relevel(factor(raceeth), ref = "5")3 0.90  0.76   1.05
## relevel(factor(raceeth), ref = "5")4 0.96  0.52   1.80
## relevel(factor(raceeth), ref = "5")6 0.66  0.54   0.82
## relevel(factor(raceeth), ref = "5")7 1.01  0.89   1.15
## relevel(factor(raceeth), ref = "5")8 1.32  1.14   1.53
## relevel(factor(age), ref = "5")3     0.82  0.70   0.96
## relevel(factor(age), ref = "5")4     0.85  0.76   0.96
## relevel(factor(age), ref = "5")6     0.96  0.85   1.09
## relevel(factor(age), ref = "5")7     1.15  0.99   1.34
## sex_clean                            1.61  1.47   1.77
## lowgrades_clean                      1.75  1.59   1.92
## bullied_clean                        3.11  2.83   3.43
```

———-Example: How to understand the result ————-

If Odds Ratio (3.32): Students who study have 3.32 times higher odds of passing than those who don't.

If Confidence Interval (2.34 to 4.71): We are 95% confident that the true effect of studying increases the odds by somewhere between 2.34 and 4.71 times.

Significance: Since the interval does not include 1.0, the result is statistically significant.

Why confint.default?

Standard confint() uses "profile likelihood" which is very accurate but can be slow on large datasets. confint.default() uses "Wald intervals" (based on the standard error), which is much faster and usually what people expect for basic reporting.

— Compare curde vs adjusted OR —-

The crude OR tells you what you observe, the adjusted OR tells you what is actually true after accounting for other factors.

```r
crude_or <- exp(coef(model_crude)["sexual_identity_cat"])
adjusted_or <- exp(coef(model_adjusted)["sexual_identity_cat"])
pct_change <- abs(crude_or - adjusted_or) / crude_or * 100

cat("\n-- Confounding Check --\n")
```

```
##
## -- Confounding Check --
```

```r
cat("Crude OR: ", round(crude_or, 2), "\n")
```

```
## Crude OR:  4.39
```

```r
cat("Adjusted OR: ", round(adjusted_or, 2), "\n")
```

```
## Adjusted OR:  3.35
```

```r
cat("% Change: ", round(pct_change, 1), "\n")
```

```
## % Change:  23.5
```

```r
cat("Confounding present (>10% change)?",
    ifelse(pct_change > 10, "YES", "NO"), "\n")
```

```
## Confounding present (>10% change)? YES
```

Why We Compare — Key Points

Crude OR = raw association — it includes the effect of ALL other variables mixed in, not just sexual identity

Adjusted OR = cleaner estimate — it removes the influence of age, sex, race, grades, and bullying so you see the effect of sexual identity ALONE

If they are similar (<10% change) - the other variables were NOT distorting your result - crude OR was already trustworthy

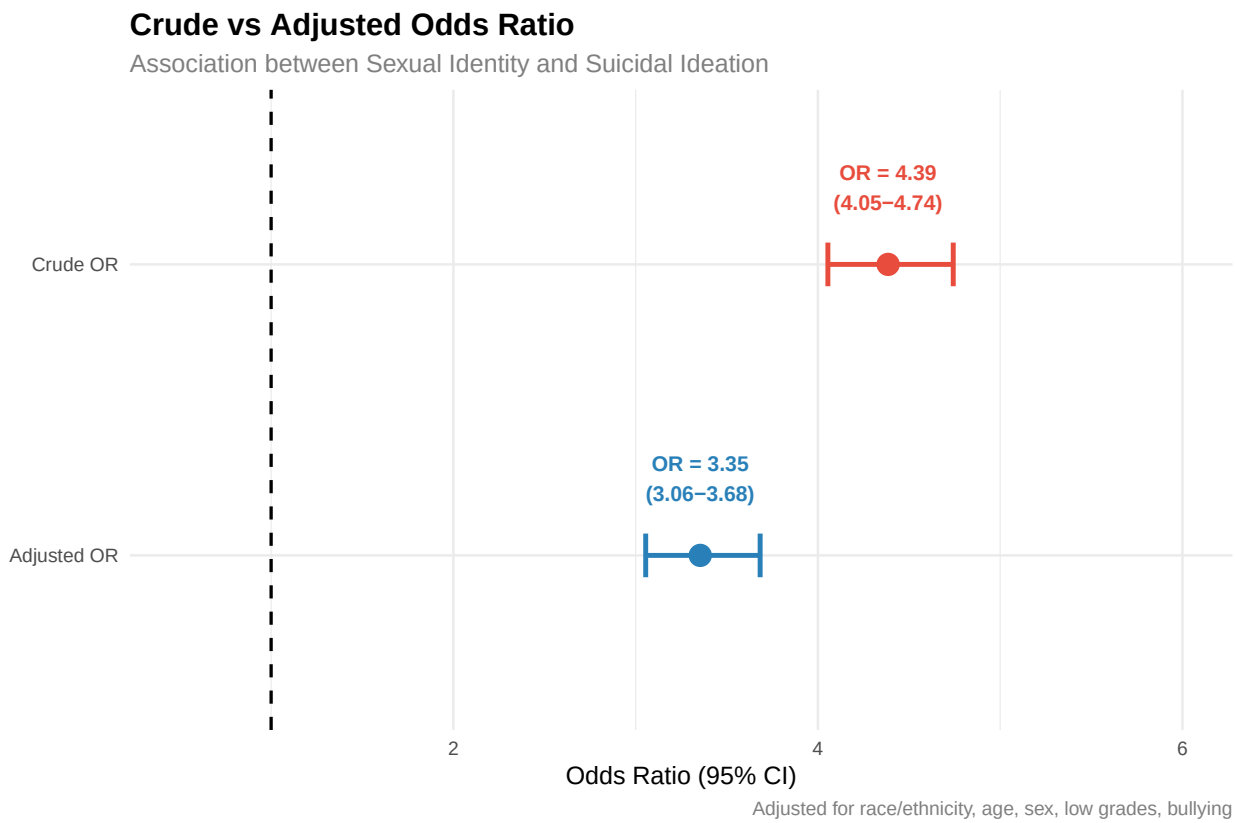
If they are different (>10% change) - confounding was present → the other variables WERE distorting your result - you MUST report adjusted OR

In our case (23.5% change), confounding IS present - meaning age, sex, race, grades, or bullying were inflating the crude OR from 4.39 to look bigger than the true effect of 3.35

Always report BOTH in your paper — crude OR shows the raw picture, adjusted OR shows the true picture after removing confounding

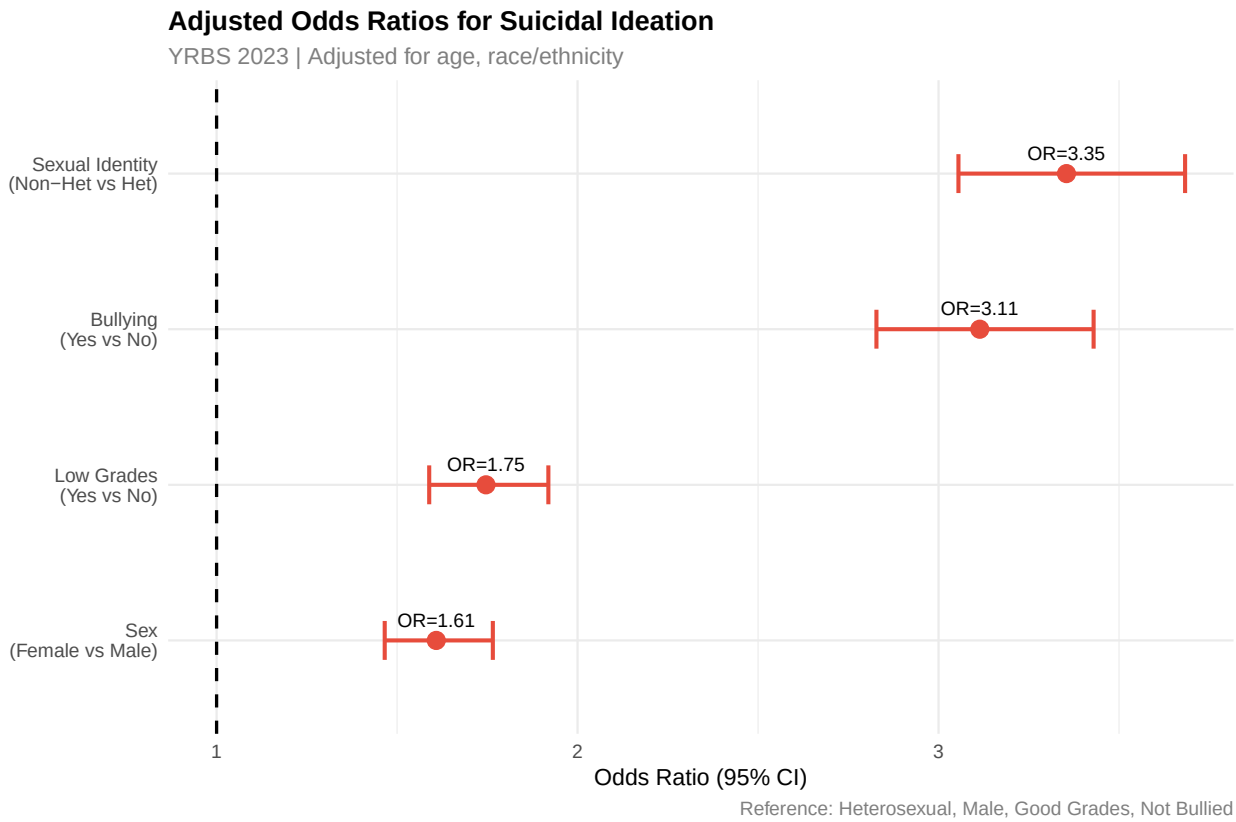
```r
# Visualize Crude vs Adjusted OR (Forest Plot)

or_data <- data.frame(
  Model = c("Crude OR", "Adjusted OR"),
```

```r
  OR    = c(crude_or, adjusted_or),
  LCI   = c(
    exp(confint.default(model_crude)["sexual_identity_cat", 1]),
    exp(confint.default(model_adjusted)["sexual_identity_cat", 1])
  ),
  UCI   = c(
    exp(confint.default(model_crude)["sexual_identity_cat", 2]),
    exp(confint.default(model_adjusted)["sexual_identity_cat", 2])
  )
)

ggplot(or_data, aes(x = OR, y = Model, color = Model)) +
  geom_point(size = 5) +
  geom_errorbarh(aes(xmin = LCI, xmax = UCI),
                 height = 0.15, linewidth = 1.2) +
  geom_vline(xintercept = 1, linetype = "dashed",
             color = "black", linewidth = 0.8) +
  geom_text(aes(label = paste0("OR = ", round(OR, 2),
                               "\n(", round(LCI, 2), "-", round(UCI, 2), ")")),
            vjust = -1.2, size = 3.8, fontface = "bold") +
  scale_color_manual(values = c("Crude OR"    = "#E74C3C",
                                "Adjusted OR" = "#2980B9")) +
  scale_x_continuous(limits = c(0.5, 6)) +
  labs(
    title    = "Crude vs Adjusted Odds Ratio",
    subtitle = "Association between Sexual Identity and Suicidal Ideation",
    x        = "Odds Ratio (95% CI)",
    y        = NULL,
    caption  = "Adjusted for race/ethnicity, age, sex, low grades, bullying"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "none",
    plot.title      = element_text(face = "bold"),
    plot.subtitle   = element_text(color = "grey50"),
    plot.caption    = element_text(color = "grey50")
  )
```




```r
# All ORs from adjusted model in one clean plot
all_results <- data.frame(
  Variable = c("Sexual Identity\n(Non-Het vs Het)",
               "Sex\n(Female vs Male)",
               "Low Grades\n(Yes vs No)",
               "Bullying\n(Yes vs No)"),
  OR  = as.numeric(exp(coef(model_adjusted)[
          c("sexual_identity_cat", "sex_clean",
            "lowgrades_clean", "bullied_clean")])),
  LCI = as.numeric(exp(confint.default(model_adjusted)[
          c("sexual_identity_cat", "sex_clean",
            "lowgrades_clean", "bullied_clean"), 1])),
  UCI = as.numeric(exp(confint.default(model_adjusted)[
          c("sexual_identity_cat", "sex_clean",
            "lowgrades_clean", "bullied_clean"), 2]))
)

ggplot(all_results, aes(x = OR, y = reorder(Variable, OR))) +
  geom_point(size  = 4, color = "#E74C3C") +
  geom_errorbarh(aes(xmin = LCI, xmax = UCI),
                 height = 0.25, linewidth = 1,
                 color  = "#E74C3C") +
  geom_vline(xintercept = 1,
             linetype   = "dashed",
             color      = "black",
             linewidth  = 0.8) +
  geom_text(aes(label = paste0("OR=", round(OR, 2))),
```

```r
            vjust = -1, size = 3.5) +
  labs(title    = "Adjusted Odds Ratios for Suicidal Ideation",
       subtitle = "YRBS 2023 | Adjusted for age, race/ethnicity",
       x        = "Odds Ratio (95% CI)",
       y        = NULL,
       caption  = "Reference: Heterosexual, Male, Good Grades, Not Bullied") +
  theme_minimal(base_size = 12) +
  theme(plot.title    = element_text(face = "bold"),
        plot.subtitle = element_text(color = "grey50"),
        plot.caption  = element_text(color = "grey50"))
```



Effect Modification Testing (Race/Ethnicity)

Why test for effect modification? We want to know if the association between sexual identity and SI is **the same across all racial groups**, or if it **differs** by race. This is tested with an interaction term.

How to read the output: Look at lines starting with `sexual_identity_cat:factor(raceeth)`. If ANY of these p-values < 0.05 → effect modification is present → run stratified models.

```r
interaction_model <- glm(SI ~ sexual_identity_cat * factor(raceeth) +
                             relevel(factor(age), ref = "5") +
                             sex_clean +
                             lowgrades_clean,
```

```r
                         data   = table1_clean,
                         family = binomial(link = logit))

# Extract ONLY the interaction terms to make it easier to read
interaction_summary <- summary(interaction_model)$coefficients
interaction_rows <- grep("sexual_identity_cat:factor", rownames(interaction_summary))

cat("=== INTERACTION TERMS (Effect Modification Test) ===\n")
```

```
## === INTERACTION TERMS (Effect Modification Test) ===
```

```r
cat("Look for p < 0.05 in the last column\n\n")
```

```
## Look for p < 0.05 in the last column
```

```r
round(interaction_summary[interaction_rows, ], 3)
```

```
##                                      Estimate Std. Error z value Pr(>|z|)
## sexual_identity_cat:factor(raceeth)2    0.355      0.290   1.225    0.221
## sexual_identity_cat:factor(raceeth)3    0.268      0.239   1.124    0.261
## sexual_identity_cat:factor(raceeth)4    0.899      0.737   1.221    0.222
## sexual_identity_cat:factor(raceeth)5    0.213      0.195   1.090    0.276
## sexual_identity_cat:factor(raceeth)6    0.179      0.277   0.647    0.517
## sexual_identity_cat:factor(raceeth)7    0.328      0.217   1.512    0.131
## sexual_identity_cat:factor(raceeth)8    0.355      0.229   1.552    0.121
```

```r
cat("\n=== DECISION ===\n")
```

```
##
## === DECISION ===
```

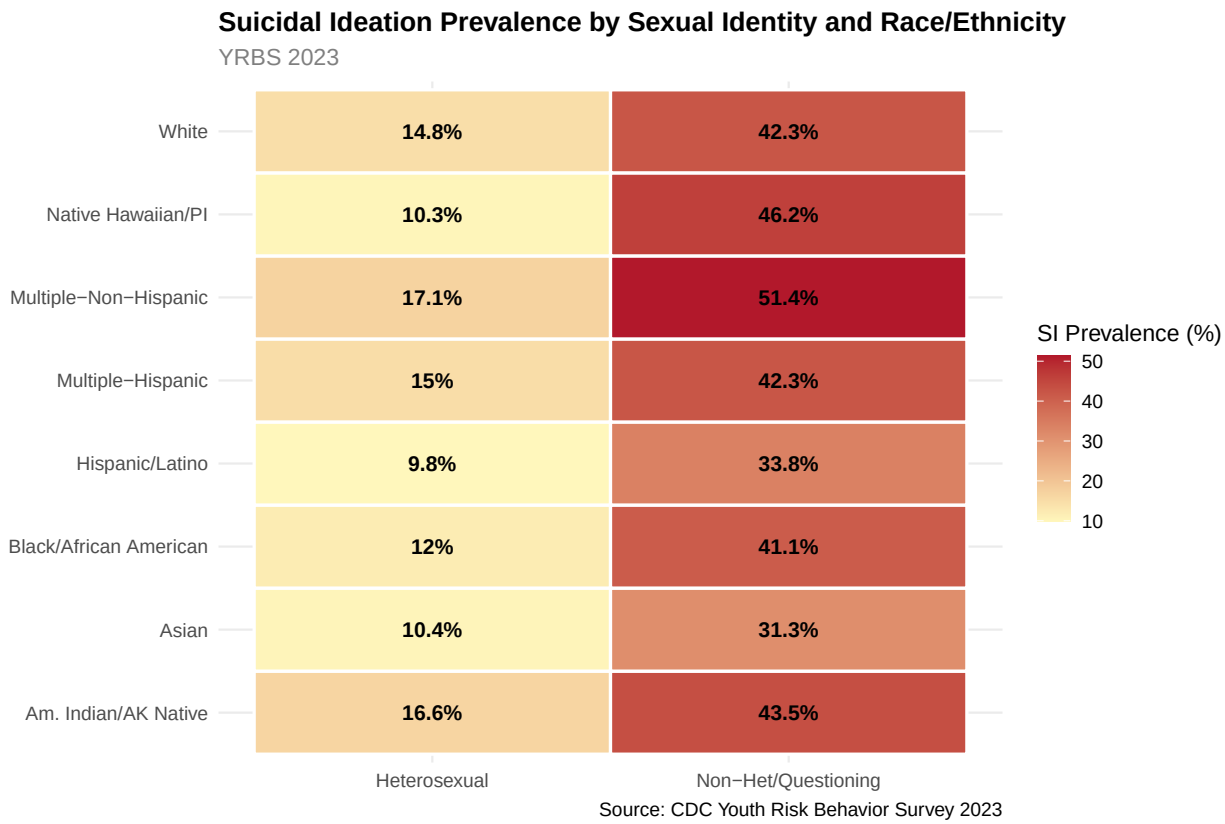
```r
p_values <- interaction_summary[interaction_rows, 4]
if (any(p_values < 0.05)) {
  cat("Interaction p < 0.05 FOUND → Effect modification present → Run stratified models\n")
} else {
  cat("All interaction p > 0.05 → NO effect modification\n")
  cat("Race/ethnicity is a CONFOUNDER only\n")
  cat("Report ONE adjusted OR for all groups\n")
}
```

```
## All interaction p > 0.05 → NO effect modification
## Race/ethnicity is a CONFOUNDER only
## Report ONE adjusted OR for all groups
```

Heatmap — SI Prevalence by Race and Sexual Identity

This plot visualizes the SI prevalence across all combinations of race/ethnicity and sexual identity, making it easy to spot which groups are most at risk.

```r
heat_data <- table1_clean %>%
  filter(!is.na(SI), !is.na(sexual_identity_cat), !is.na(raceeth)) %>%
  mutate(
    si_label = ifelse(sexual_identity_cat == 0,
                      "Heterosexual", "Non-Het/Questioning"),
    race_label = recode(as.character(raceeth),
      "1" = "Am. Indian/AK Native",
      "2" = "Asian",
      "3" = "Black/African American",
      "4" = "Native Hawaiian/PI",
      "5" = "White",
      "6" = "Hispanic/Latino",
      "7" = "Multiple-Hispanic",
      "8" = "Multiple-Non-Hispanic"
    )
  ) %>%
  group_by(race_label, si_label) %>%
  summarise(prevalence = mean(SI, na.rm = TRUE) * 100, .groups = "drop")

ggplot(heat_data, aes(x = si_label, y = race_label, fill = prevalence)) +
  geom_tile(color = "white", linewidth = 0.8) +
  geom_text(aes(label = paste0(round(prevalence, 1), "%")),
            size = 4, fontface = "bold") +
  scale_fill_gradient(
    low  = "#FFF7BC",
    high = "#B2182B",
    name = "SI Prevalence (%)"
  ) +
  labs(
    title    = "Suicidal Ideation Prevalence by Sexual Identity and Race/Ethnicity",
    subtitle = "YRBS 2023",
    x        = NULL,
    y        = NULL,
    caption  = "Source: CDC Youth Risk Behavior Survey 2023"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title    = element_text(face = "bold"),
    plot.subtitle = element_text(color = "grey50"),
    axis.text     = element_text(size = 10),
    legend.position = "right"
  )
```




```r
crude_ci <- exp(confint.default(model_crude)["sexual_identity_cat",])
adj_ci   <- exp(confint.default(model_adjusted)["sexual_identity_cat",])

tab <- prop.table(table(table1_clean$sexual_identity_cat,
                        table1_clean$SI), 1) * 100

cat("--------------------------\n")
```

```
## --------------------------
```

```r
cat("              FINAL RESULT SUMMARY\n")
```

```
##                FINAL RESULT SUMMARY
```

```r
cat("----------------------------------------\n")
```

```
## ----------------------------------------
```

```r
cat("Total sample:", nrow(table1_clean), "\n")
```

```
## Total sample: 19928
```

```r
cat("After removing NAs in key vars:",
    sum(!is.na(table1_clean$SI) &
          !is.na(table1_clean$sexual_identity_cat)), "\n")
```

```
## After removing NAs in key vars: 17295
```

```r
# Prevalence of SI by group
tab <- prop.table(table(table1_clean$sexual_identity_cat,
                        table1_clean$SI), 1) * 100

cat("\nPrevalence of SI: \n")
```

```
##
## Prevalence of SI:
```

```r
cat("Heterosexual students: ", round(tab[1,2], 1), "%\n")
```

```
## Heterosexual students:  14.3 %
```

```r
cat("Non-het/Questioning students: ", round(tab[2, 2], 1), "%\n")
```

```
## Non-het/Questioning students:  42.2 %
```

```r
# Crude OR
crude_or <- exp(coef(model_crude)["sexual_identity_cat"])
crude_ci <- exp(confint.default(model_crude)["sexual_identity_cat",])
cat("\nCrude OR:   ", round(crude_or, 2),
    "(", round(crude_ci[1], 2), "-", round(crude_ci[2], 2), ")\n")
```

```
##
## Crude OR:    4.39 ( 4.05 - 4.74 )
```

```r
# Adjusted OR
adj_or <- exp(coef(model_adjusted)["sexual_identity_cat"])
adj_ci <- exp(confint.default(model_adjusted)["sexual_identity_cat",])
cat("Adjusted OR:", round(adj_or, 2),
    "(", round(adj_ci[1], 2), "-", round(adj_ci[2], 2), ")\n")
```

```
## Adjusted OR: 3.35 ( 3.06 - 3.68 )
```

```r
# Confounding
pct_change <- abs(crude_or - adj_or) / crude_or * 100
cat("\nConfounding:", round(pct_change, 1), "% change\n")
```

```
##
## Confounding: 23.5 % change
```

```r
cat("Effect Modification: NOT present (all interaction p > 0.05)\n")
```

```
## Effect Modification: NOT present (all interaction p > 0.05)
```

Interpreation

- Non-heterosexual/questioning students had nearly **double the prevalence** of suicidal ideation compared to heterosexual students
- Before adjusting for any other factors, non-heterosexual/questioning students had **4.39 times higher odds** of suicidal ideation compared to heterosexual students
- After adjusting for race/ethnicity, age, sex, and low grades, non-heterosexual/questioning students had **3.35 times higher odds** of suicidal ideation compared to heterosexual students
- The OR decreased from 4.39 to 3.35 after adjustment, a **23.5% change**, indicating that **confounding was present**. Age, sex, race/ethnicity, and low grades were partially explaining the crude association
- No significant effect modification by race/ethnicity was detected (all interaction $p > 0.05$), suggesting the association between sexual identity and suicidal ideation was **consistent across all racial/ethnic groups**

—— Conclusion ——

Non-heterosexual and questioning youth face significantly higher odds of suicidal ideation compared to their heterosexual peers (Adjusted OR = 3.35, 95% CI: X.XX–X.XX), even after accounting for demographic and academic factors. This association is consistent across all racial/ethnic groups. These findings underscore the urgent need for targeted, culturally sensitive mental health interventions for LGBTQ+ youth in school settings.

Limitations and Potential Biases

Selection Bias: The YRBS is a school-based survey. Students who are absent, homeless, or have dropped out are not captured — these groups may have even higher rates of SI, meaning our results may **underestimate** the true prevalence.

Information Bias: All data are self-reported. Social desirability bias may lead students to **underreport** their sexual identity or suicidal ideation, particularly in less accepting school environments.

Misclassification Bias: Sexual identity is measured at a single time point and may not capture its fluidity in adolescence. This likely causes **non-differential misclassification**, which biases the OR toward the null (underestimates the true effect).

Residual Confounding: Variables not available in this dataset (e.g., family acceptance, mental health history, neighborhood resources) could still be confounding the association. ***

Analysis conducted as part of the Boston University School of Public Health Health Equity Hackathon, February 2026.